

Mehrdad Momeni zadeh

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TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C++, Go, SQL

Frameworks: Spring Boot 3, Hibernate, NestJS, Next.js, React, Django, ShadCN

Libraries: PyTorch, scikit-learn, TensorFlow, WandB, Hydra, OpenCV, FFmpeg

DevOps & Cloud: Docker, Kubernetes, Maven, Gradle, Jenkins, GCP, AWS (S3), Git, JWT

Database & Storage: PostgreSQL, MySQL, MongoDB, Firebase, H2

WORK EXPERIENCE

DeLTA Lab — Simon Fraser University

Burnaby, BC

Undergraduate Researcher — Scientific Machine Learning

Dec 2024 – May 2025

- **Second author** on **NeurIPS 2025** paper on data-efficient neural operators.
- Boosted PDE benchmark accuracy by **12%** and cut training cost **4×** using Google Cloud TPU-v4.
- Added **rotary positional embeddings** for stable multi-step forecasting without drift.
- Developed 1-command experiment runners (**Hydra + W&B**) enabling teammates to launch tests in **<10 min**.
- Containerised the research stack with **Docker** for reproducible deployments across labs.

TECHNICAL PROJECTS

StagePass — Full-stack event ticketing system for university events.

[GitHub](#) 📄

Tech: Java, Spring Boot 3, Spring Security, OAuth2, PostgreSQL, React, REST API

- Deployed at **SFU's 2025 FASS Formal**, processing tickets for **100+ students**.
- Secured organizer/attendee workflows with Spring Security and OAuth2.
- Designed REST APIs and PostgreSQL schema for event creation, ticket sales, and validation.

ShareScreenly — Full-stack platform for screen recording and instant video sharing.

[GitHub](#) 📄

Tech: TypeScript, Next.js 15, Node.js, PostgreSQL, Tailwind CSS, S3, Better Auth, FFmpeg

- Adopted by **100+ SFU students** for quick demo sharing, reducing setup time from **5 min → 30 sec**.
- Integrated AWS S3 + FFmpeg for scalable video storage, compression, and metadata handling.
- Implemented secure role-based access with Better Auth and built a responsive Tailwind UI.

MicroGopher — Distributed microservices platform in Go for scalable systems.

[GitHub](#) 📄

Tech: Go, Docker, Kubernetes, PostgreSQL, MongoDB, gRPC, AMQP

- Split a monolith into modular Go microservices, improving fault isolation and performance.
- Built services for auth, logging, and mailer communicating via REST and gRPC.
- Deployed on Docker Swarm + Kubernetes, sustaining **10K+ requests/sec** with minimal downtime.

HirePrep — AI-powered mock interview platform with real-time, voice-based practice.

[GitHub](#) 📄

Tech: TypeScript, Next.js 15, Tailwind CSS, Firebase, Gemini AI, Vapi, ShadCN, Zod

- Piloted with **50+ SFU students**, improving interview performance scores by **20%**.
- Implemented Firebase + Firestore for secure account/session management.
- Generated tailored interview questions and feedback using Gemini AI.

NeuroDriver — 2D driving simulator with neural agents trained via genetic algorithms.

[GitHub](#) 📄

Tech: Python, Box2D, Genetic Algorithms, Makefile, NumPy

- Trained neural agents in a Box2D simulator, cutting collisions **20%** in **50** generations.
- Boosted lap completion **30%** with a multi-factor reward system (speed, safety, lane discipline).
- Automated hyper-parameter sweeps and GIF replays via one-command Makefile pipeline.

EDUCATION

Simon Fraser University

Burnaby, BC

Bachelor of Science in Computer Science

GPA: 3.4 / 4.33

Expected Graduation

Dec 2026